GYMWIZARD

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ISPW

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1. Software Requirement Specification

1.1 Introduction

1.1.1 Aim of the document

The purpose of this document is to present the fundamental and specific requirements of the "GymWizard" system, a project for the Software Engineering and Web Design course, Academic Year 2024/2025.
The document provides a comprehensive overview of the system's objectives, core functionalities, technical requirements, and the rationale behind the design choices, in order to guide the development process and ensure alignment among all team members.
Prepared by: Chiara Carumani and Simone Lacroce.

1.1.2 Overview of the defined system

The system is a Java-based software platform structured according to the MVC architecture, developed with Maven, and featuring a graphical interface created using JavaFX and FXML files. The system allows for efficient and modern management of typical gym activities, including:

- Management of clients, personal trainers, and administrators.
- Management of memberships and deadlines.
- Creation and assignment of training plans.
- Viewing user progress.
- Booking of gym activities.
- Viewing reports and statistics for the sports center.
- QR code integration for gym access and use of reserved areas (e.g., hairdryers, showers, etc.).

There are three user types with different functionalities: Clients, Personal Trainers, and Administrators.

1.1.3 HW e SW requirements

Software requirements:

- Java 17+
- Scene Builder (for generating FXML files)
- Draw.io (for creating UML diagrams)
- IDE: IntelliJ IDEA or Eclipse
- MySQL Server

Hardware requirements:

- CPU: Dual-Core 2.0 GHz
- RAM: 2 GB
- Disk Space: 1 GB
- Operating System: Windows, MacOS, Linux (compatible with Java 17+)

1.1.4 Related systems

System 1: Virtuagym

Pros:

- Excellent mobile interface
- Food and fitness tracking
- Companion app

Cons:

- Complex web interface
- High cost for small centers

System 2: MyWellness (by Technogym)

Pros:

- Perfect integration with Technogym equipment
- Detailed data analysis

Cons:

- Closed and inflexible system
- Requires proprietary hardware

System 3: eGym

Pros:

- Smart access with NFC bracelets
- Highly automated gym experience

Cons:

- Not very customizable for small businesses

System 4: TeamSystem Wellness

Pros:

- Complete gym management suite
- Integration with QR code, NFC, and POS
- Tax system and CRM included
- Italian solution with dedicated support

Cons:

- High licensing cost
- Closed and inflexible architecture

1.2 User Stories

Customer:

1) As a customer, I want to renew my subscription so that I can continue going to the gym without interruptions.
2) As a customer, I want to view and book available activities so that I don't have to worry about not finding a spot.

Personal Trainer:

1) As a personal trainer, I want to edit my clients' workout plans so that I can manage their programming.
2) As a personal trainer, I want to create new workout plans so that I can assign personalized programs to my clients.

Admin:

1) As an administrator, I want to check the gym's reports and statistics so that I can analyze overall performance and make informed decisions.
2) As an administrator, I want to create new activities by setting the time and date so that I can expand the fitness center's offerings.

1.3 Functional Requirements

Activity Booking Management:

- The system shall register the booking of an activity selected by the user.
- The system shall store and display the updated list of the user's active bookings.

Subscription Management:

- The system shall display the current status of the subscription associated with the authenticated user.
- The system shall manage the renewal of the subscription by allowing the selection of one of the available options.

Training Plan Management:

- The system shall generate a new training plan for a client selected by the personal trainer.
- The system shall display the active training plan associated with the authenticated user.

Entry Code:

- The system shall generate a unique entry code and shall validate it to authorize access to the gym.

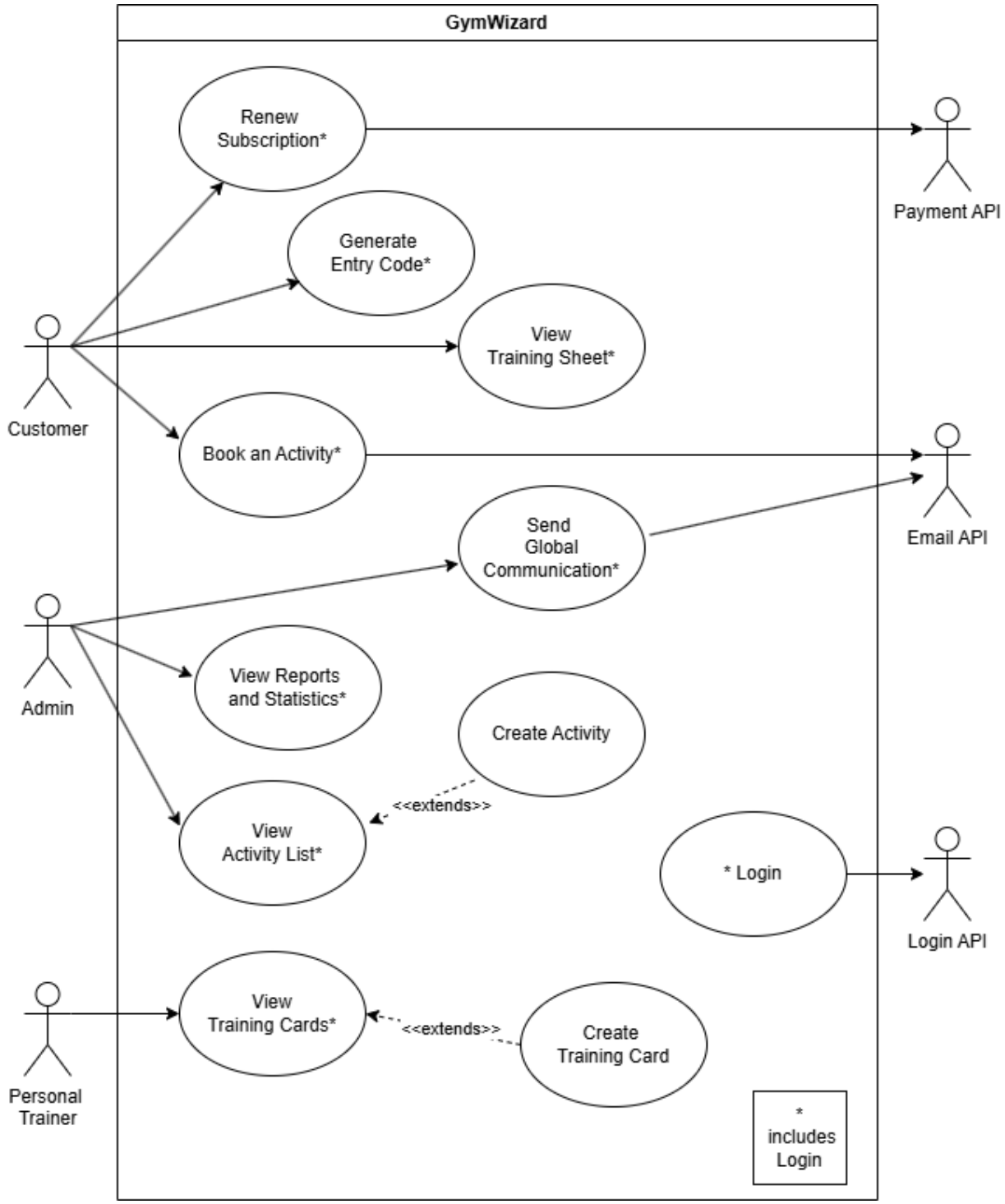
Activity Management:

- The system shall create a new activity and shall register it in the catalog of available activities.

Reports and Statistics:

- The system shall generate and display reports and statistics related to the gym's performance.

1.4 Use Cases



1.4.1 Internal Steps

Use case: Renew Subscription

Main flow:

1. The customer accesses the "Subscription Status" section.
2. The system displays the details of the current subscription.
3. The customer selects "Renew Subscription".
4. The system displays the available plans (10-month, monthly, quarterly, annual) and the price.
5. The customer selects the desired plan.
6. The system displays the summary page.
7. The customer chooses the payment method (PayPal).
8. The system processes the payment, notifies the customer, and activates the new plan.

Extensions:

3a. The customer already has an active subscription.
3b. The system displays an error.
8a. The payment fails.
8b. The system notifies the customer and allows them to retry or change the payment Method.

Use case: Book an Activity

Main flow:

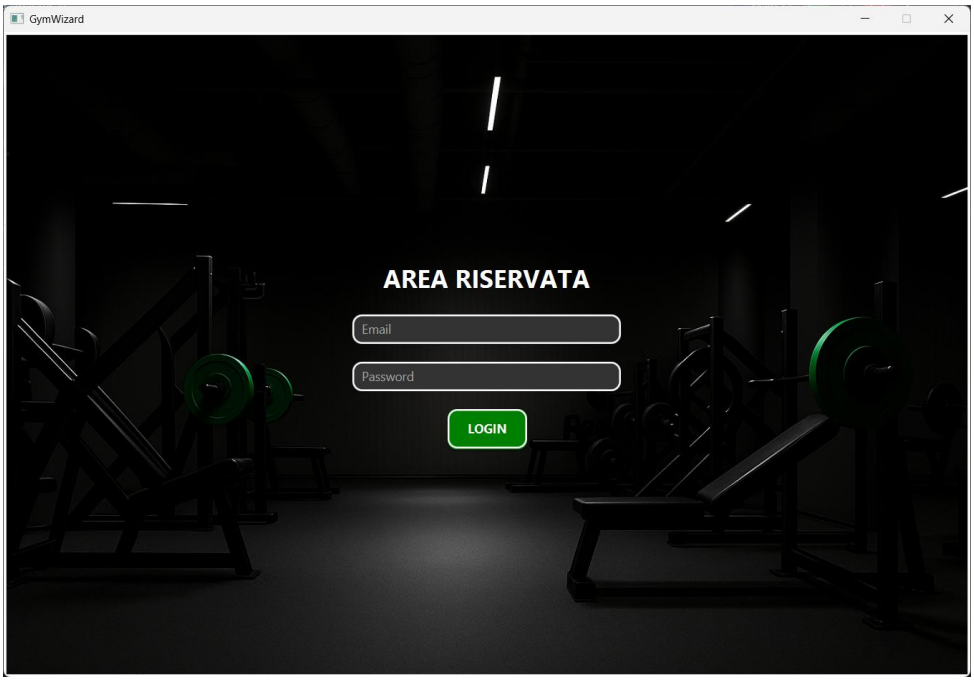
1. The customer accesses the "Activity List" section.
2. The system displays the list of available activities.
3. The customer selects the desired activity.
4. The system displays the activity summary.
5. The customer confirms the booking.
6. The system records the booking and notifies the customer.
7. The system updates the available slots for the activity.
8. The system sends a booking confirmation to the user via email.

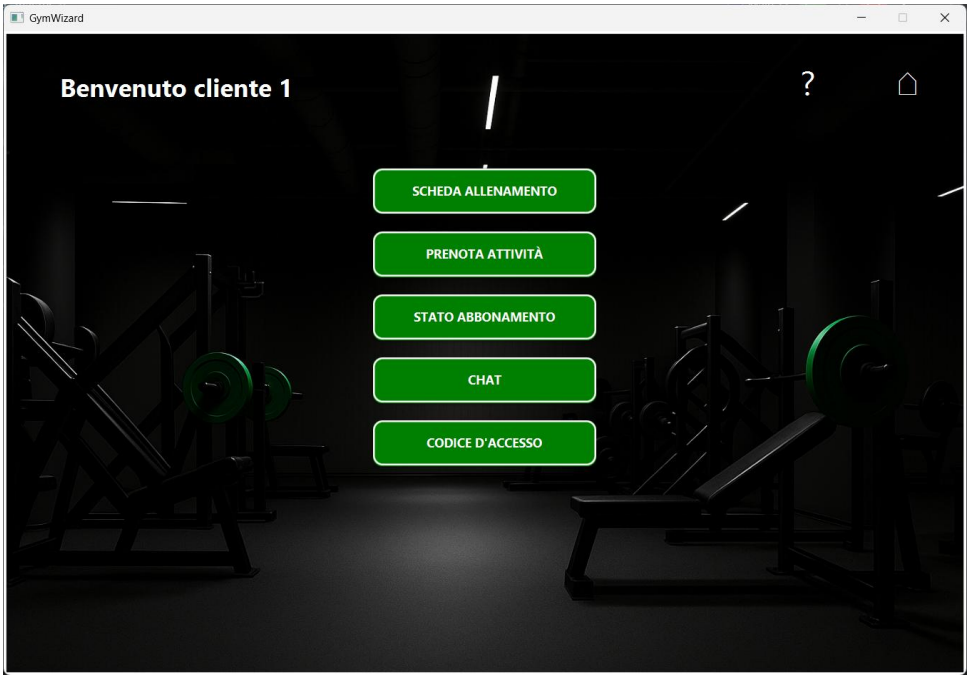
Extensions:

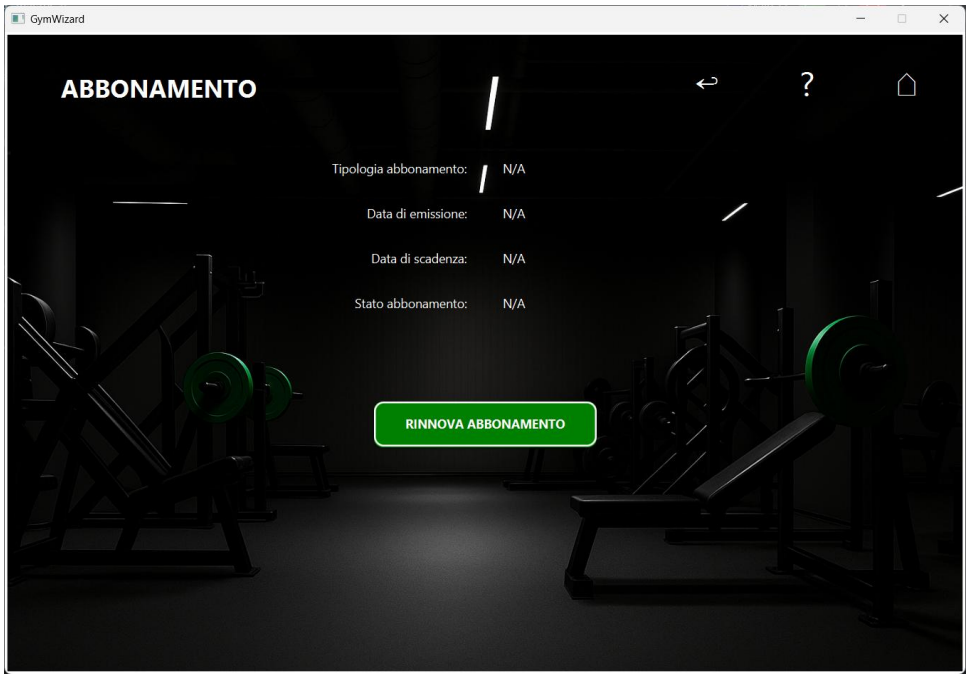
1a. The system checks the subscription status and detects that it has expired.
1b. The user is redirected to the subscription renewal screen.
4a. The system detects that there are no available slots for that activity.
4b. The system redirects to the activity selection page.

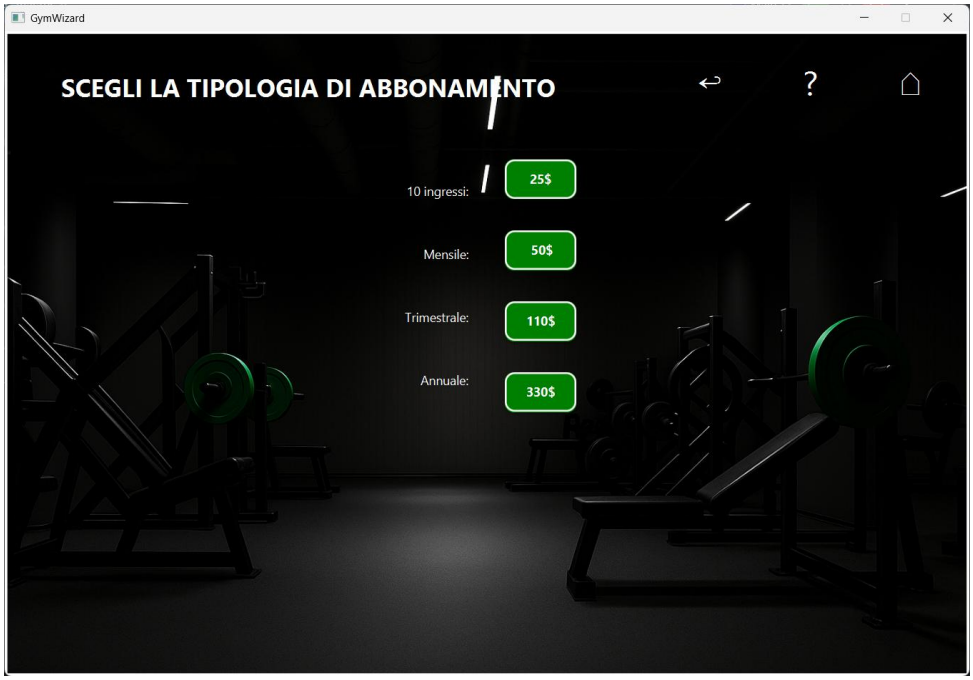
2. Storyboards

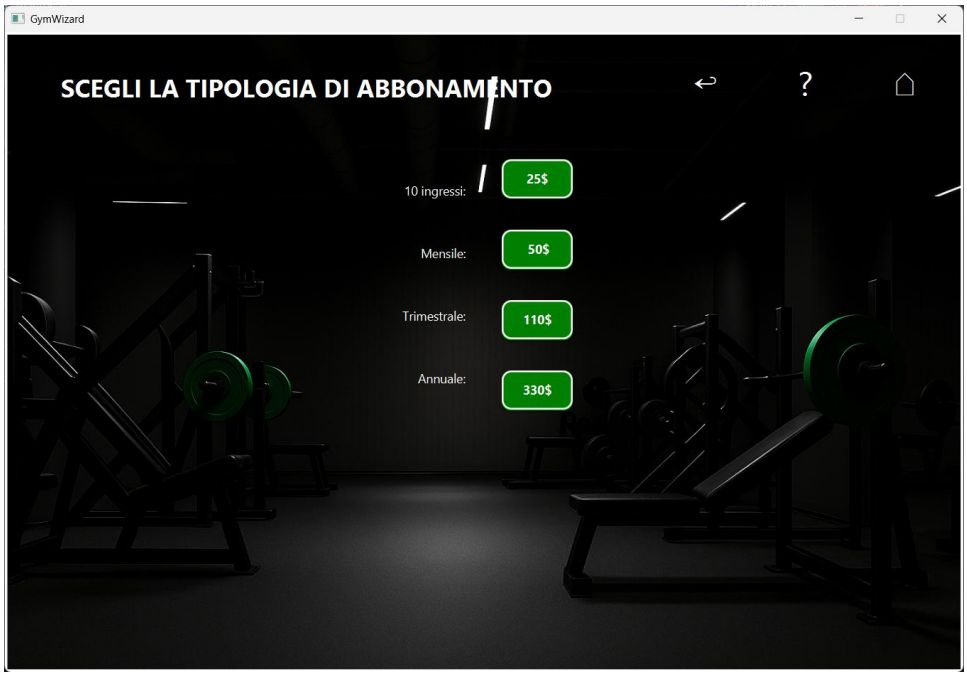
Renew Subscription

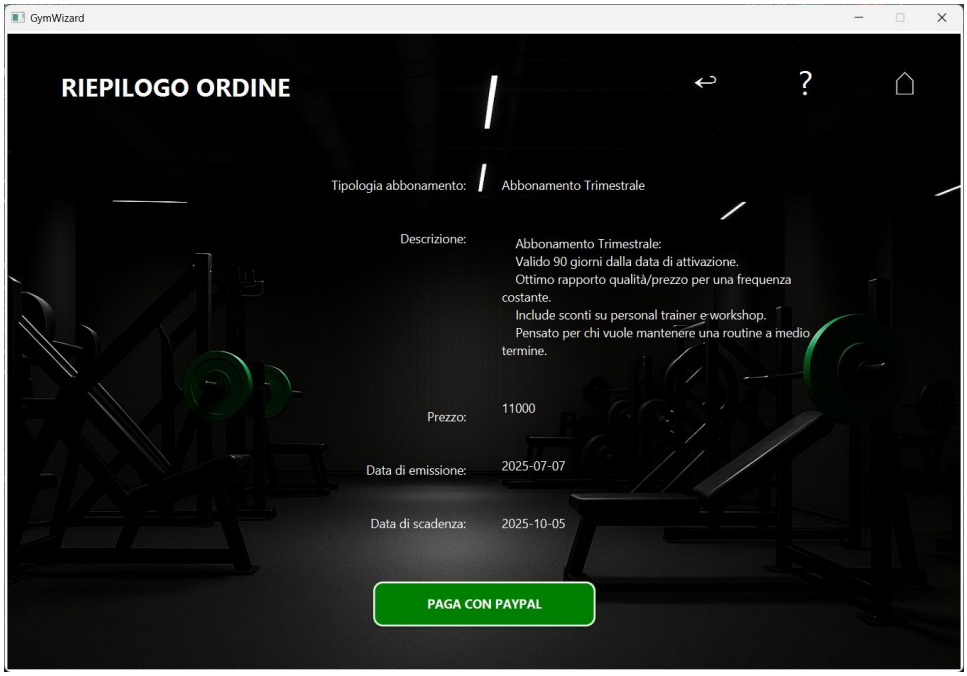

GymWizard
AREA RISERVATA
Email
Password
LOGIN


GymWizard
Benvenuto cliente 1
?
SCHEDA ALLENAMENTO
PRENOTA ATTIVITÀ
STATO ABBONAMENTO
CHAT
CODICE D'ACCESSO

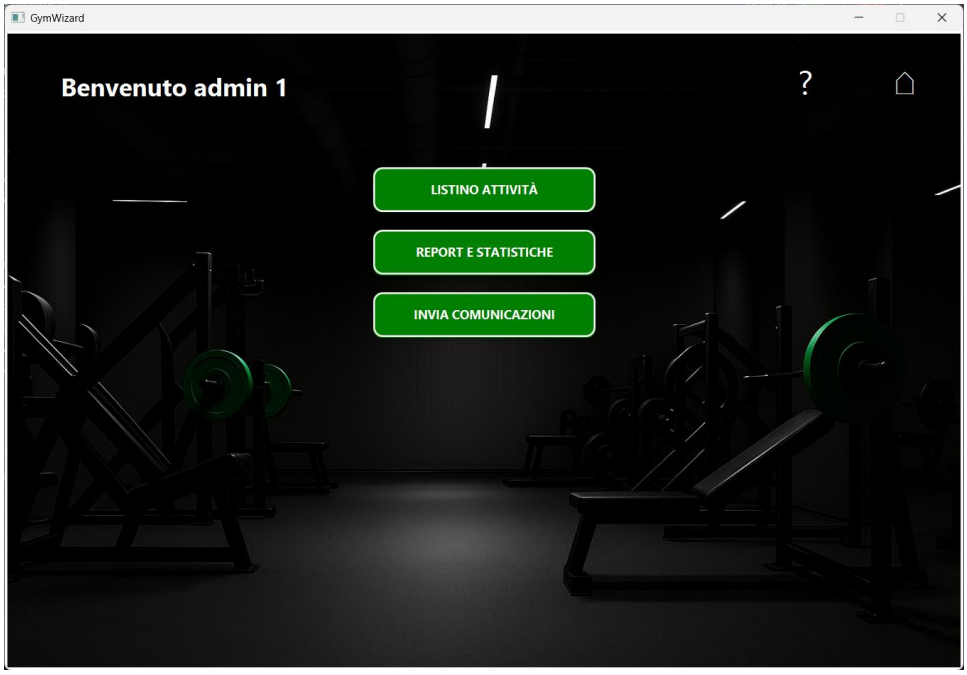

GymWizard
ABBONAMENTO
Tipologia abbonamento: N/A
Data di emissione: N/A
Data di scadenza: N/A
Stato abbonamento: N/A
RINNOVA ABBONAMENTO

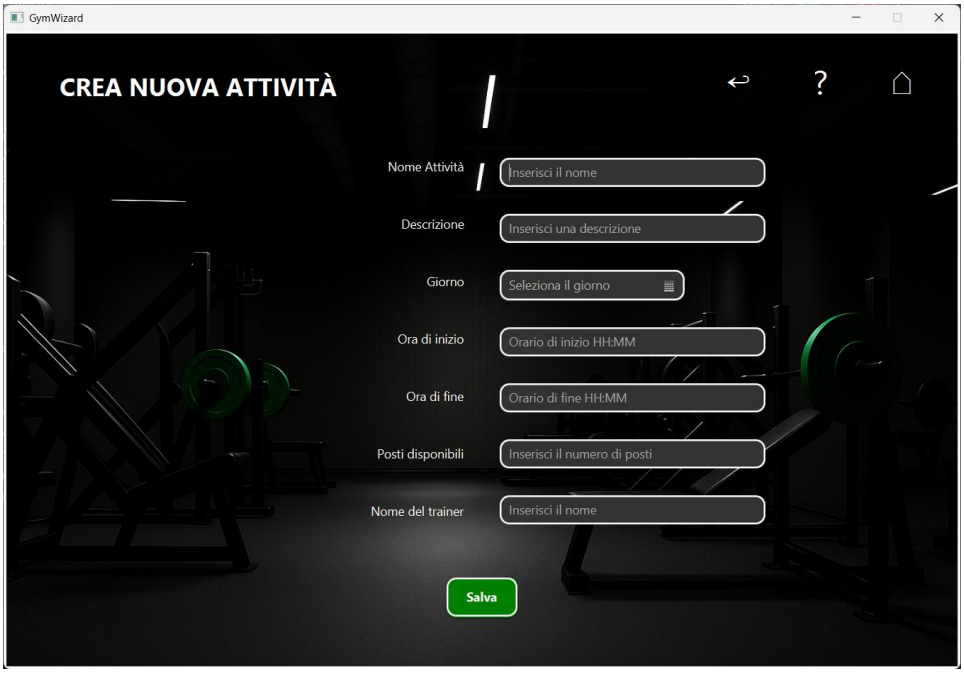

GymWizard
SCEGLI LA TIPOLOGIA DI ABBONAMENTO
10 ingressi: 25$
Mensile: 50$
Trimestrale: 110$
Annuale: 330$

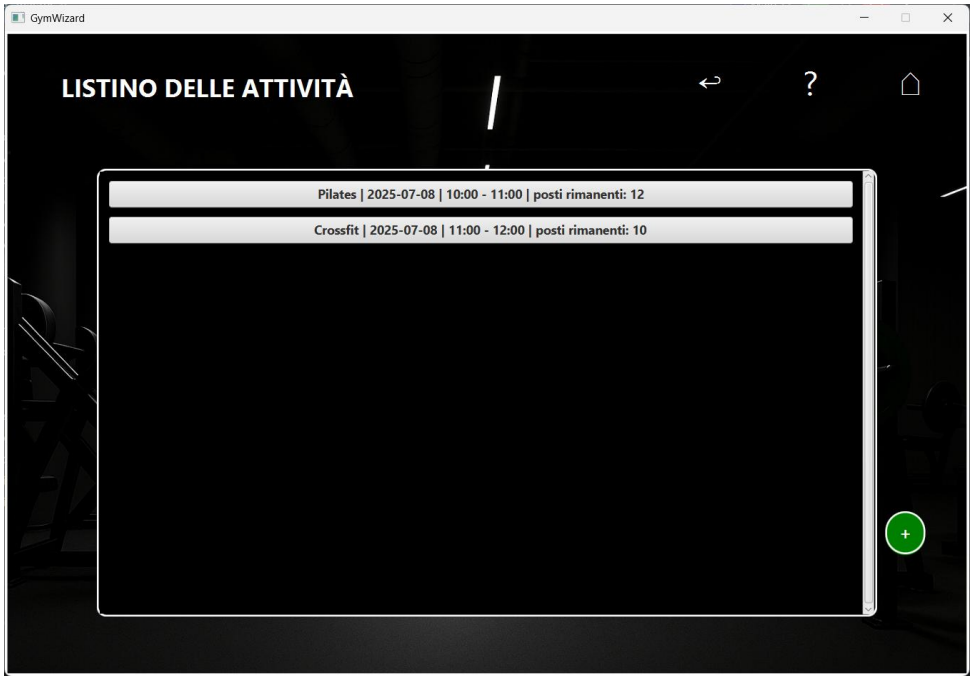

GymWizard
SCEGLI LA TIPOLOGIA DI ABBONAMENTO
10 ingressi:
25$
Mensile:
50$
Trimestrale:
110$
Annuale:
330$

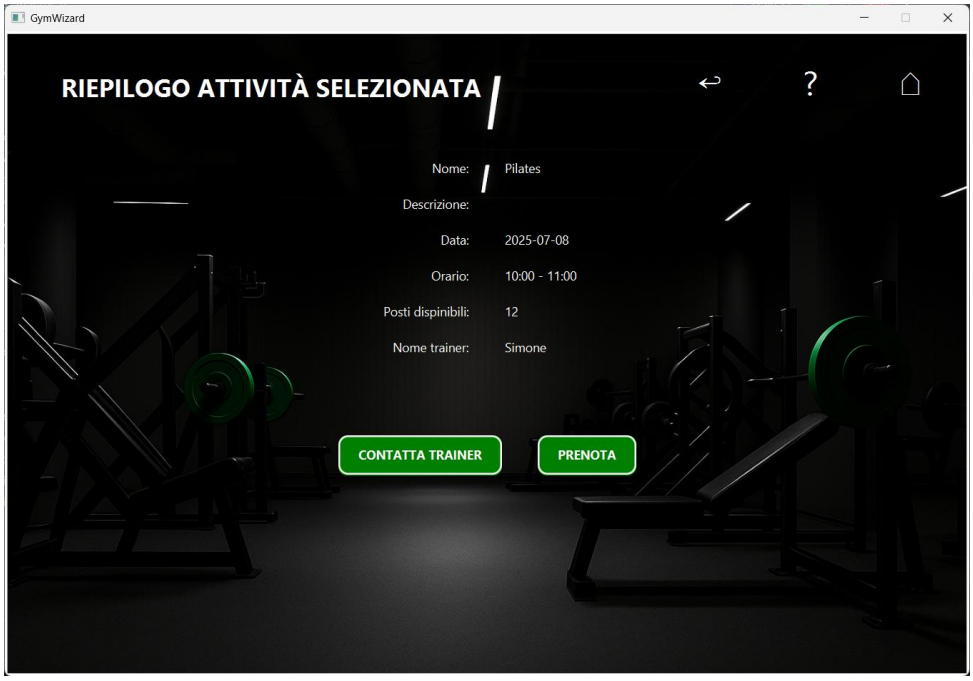

GymWizard
RIEPILOGO ORDINE
Tipologia abbonamento:
Abbonamento Trimestrale
Descrizione:
Abbonamento Trimestrale:
Valido 90 giorni dalla data di attivazione.
Ottimo rapporto qualità/prezzo per una frequenza costante.
Include sconti su personal trainer e workshop.
Pensato per chi vuole mantenere una routine a medio termine.
Prezzo:
11000
Data di emissione:
2025-07-07
Data di scadenza:
2025-10-05
PAGA CON PAYPAL

Create Activity






GymWizard
LISTINO DELLE ATTIVITÀ
Pilates | 2025-07-08 | 10:00 - 11:00 | posti rimanenti: 12
Crossfit | 2025-07-08 | 11:00 - 12:00 | posti rimanenti: 10
+

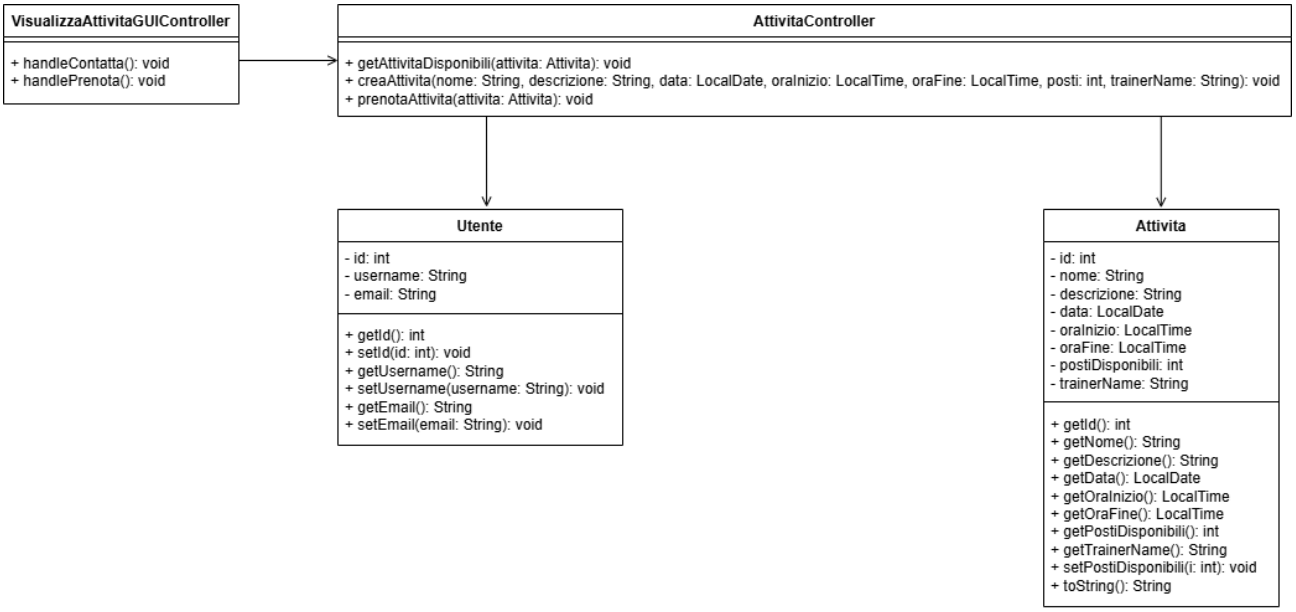

GymWizard
RIEPILOGO ATTIVITÀ SELEZIONATA
Nome: Pilates
Descrizione:
Data: 2025-07-08
Orario: 10:00 - 11:00
Posti dispinibili: 12
Nome trainer: Simone
CONTATTA TRAINER
PRENOTA

3. Design

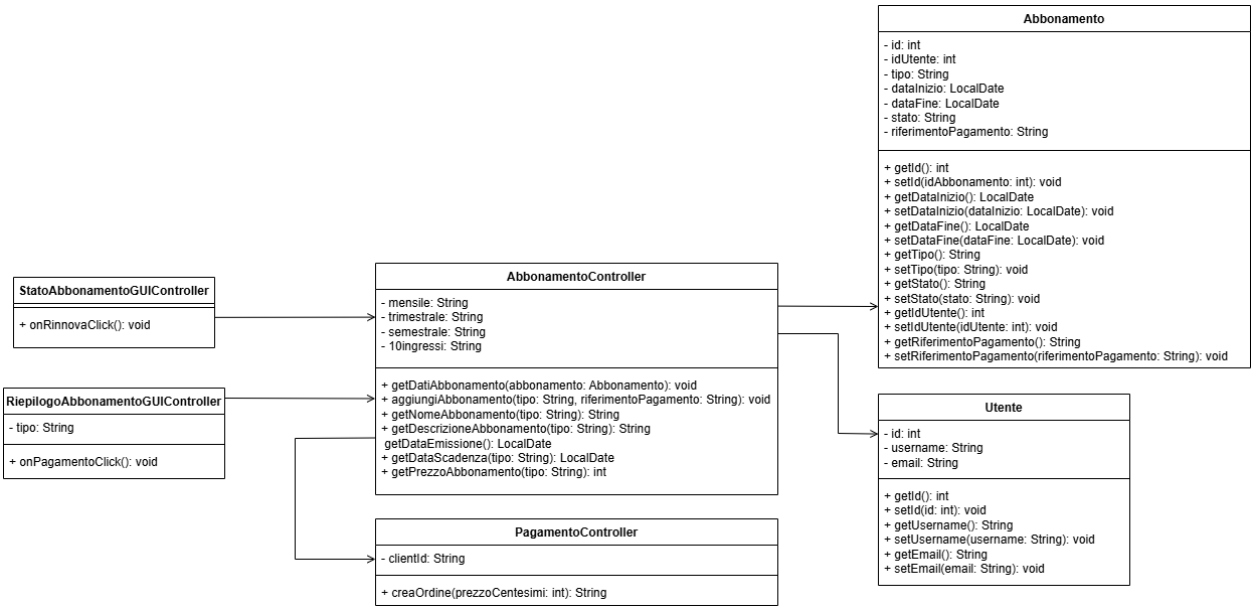
3.1 Class Diagram

3.1.1 VOPC

Book an Activity:

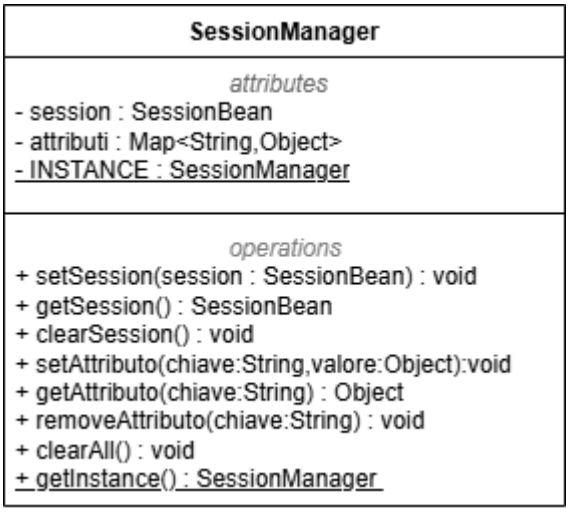


Renew Subscription:

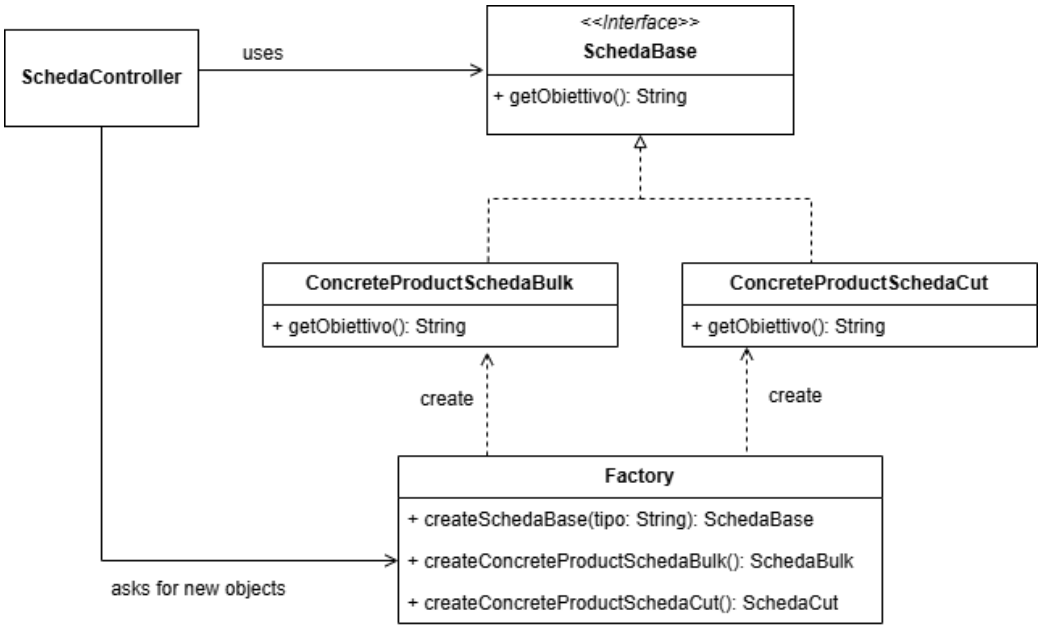


3.1.2 Design-Level Diagram

Singleton:



Factory Method:



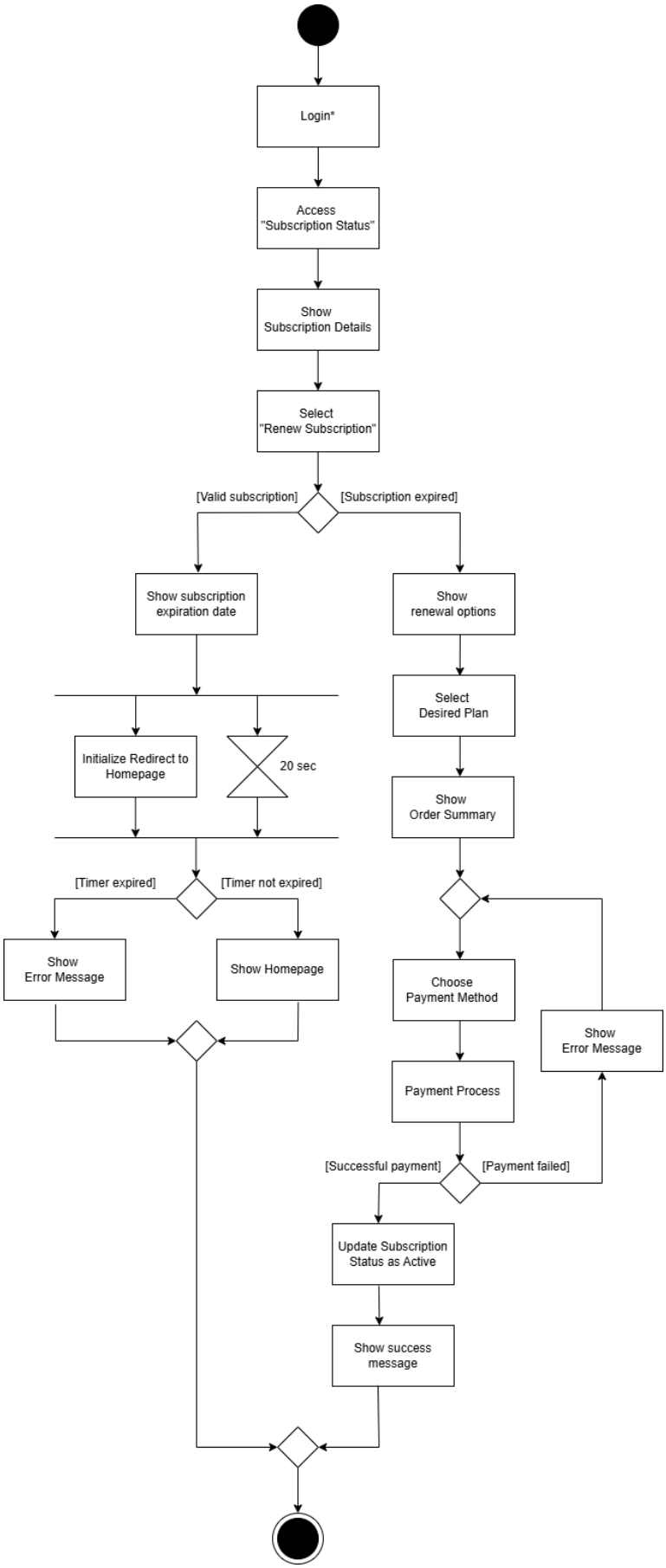
3.2 Design Patterns

Factory Method Pattern:
In GymWizard, we've applied the Factory Method Pattern to manage the creation of different workout plans through the Factory class. This approach encapsulates the instantiation logic for various types of plans (ConcreteProductSchedaBulk, ConcreteProductSchedaCut), exposing them through a common interface SchedaBase. This makes it possible to create objects based on a parameter (e.g., "bulk" or "cut") without hardcoding the specific class in the controllers. By delegating the instantiation to the Factory, we achieve greater flexibility, making it easier to add new types of workout plans in the future without modifying existing client code. This improves maintainability and keeps the codebase clean and modular.

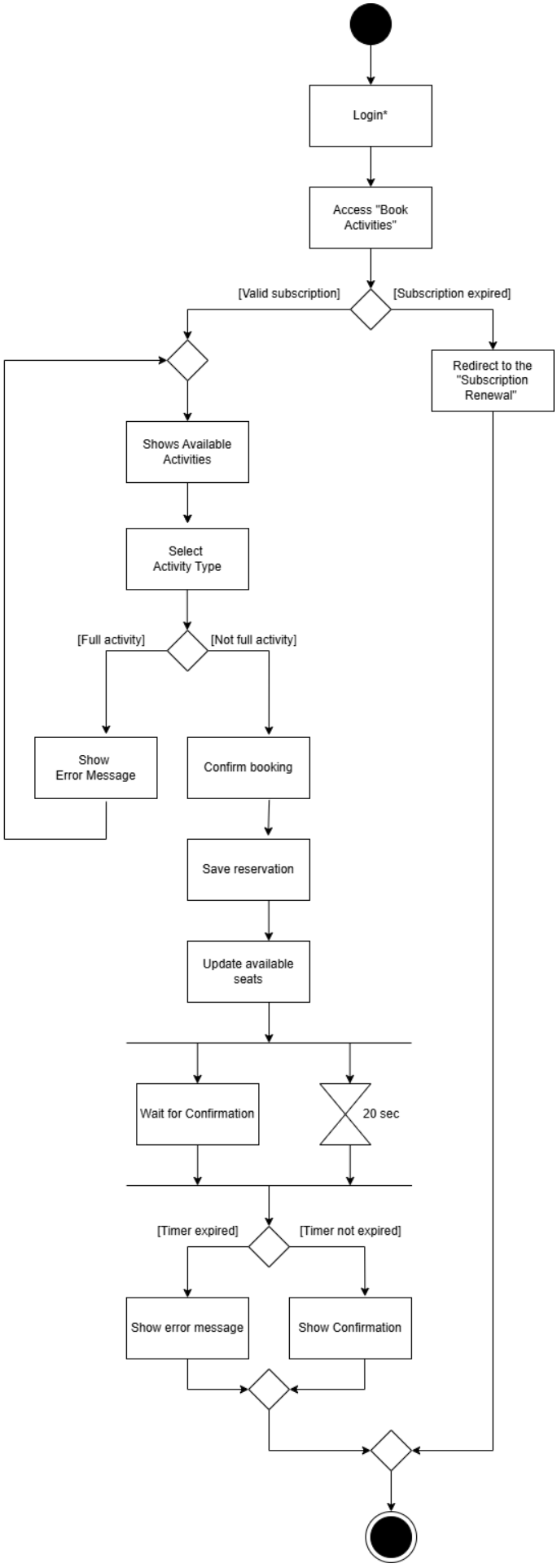
Singleton Pattern:
In GymWizard, we've adopted the Singleton Pattern for managing user sessions via the SessionManager class, maintaining a shared, centralized state throughout the app's execution. This allows us to consistently access logged-in user information from any part of the system, avoiding duplication or inconsistencies. Instance uniqueness, guaranteed by getInstance(), allows all controllers to reference the same session without explicit object transfers, making the code cleaner and more modular. Furthermore, the use of a key-value map to manage attributes makes the system flexible and easily extensible.

3.3 Activity Diagram

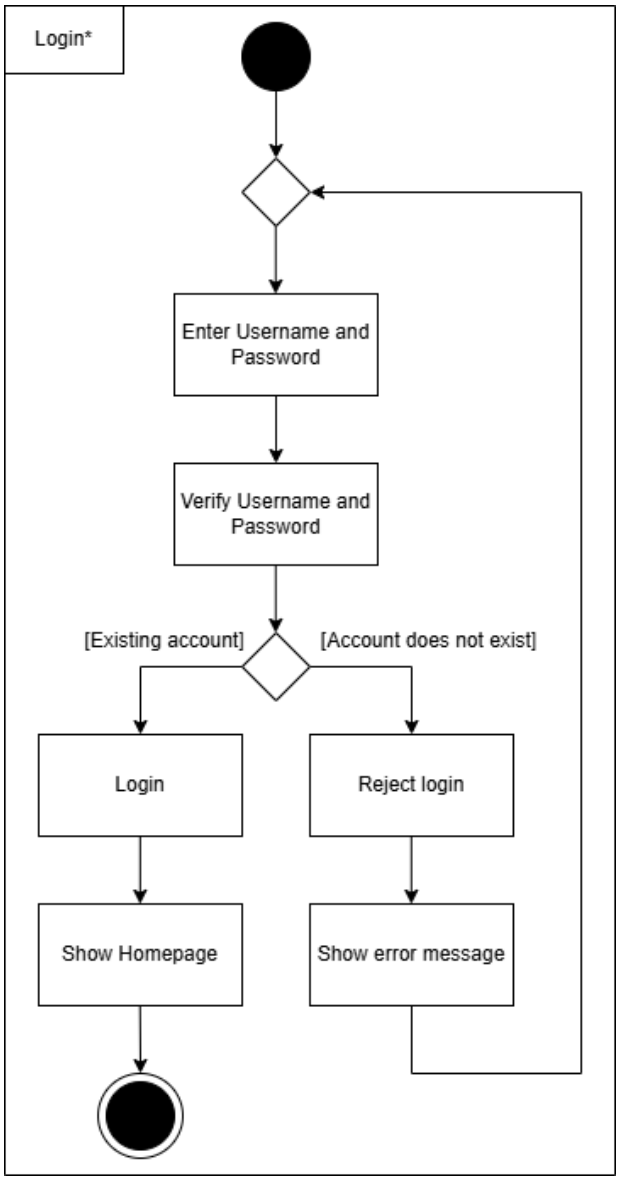
UC: Renew Subscription



UC: Book an Activity

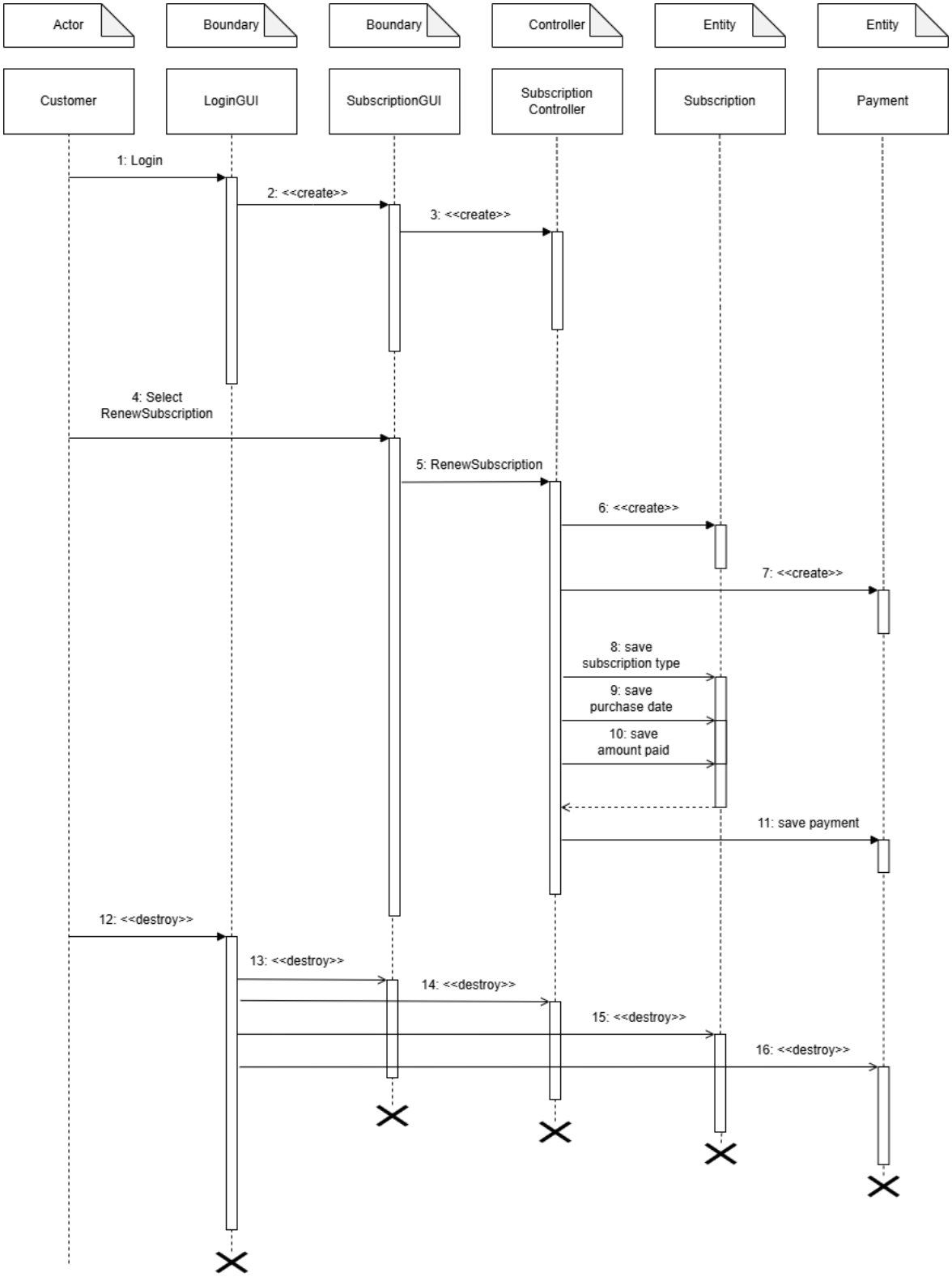


UC: Login

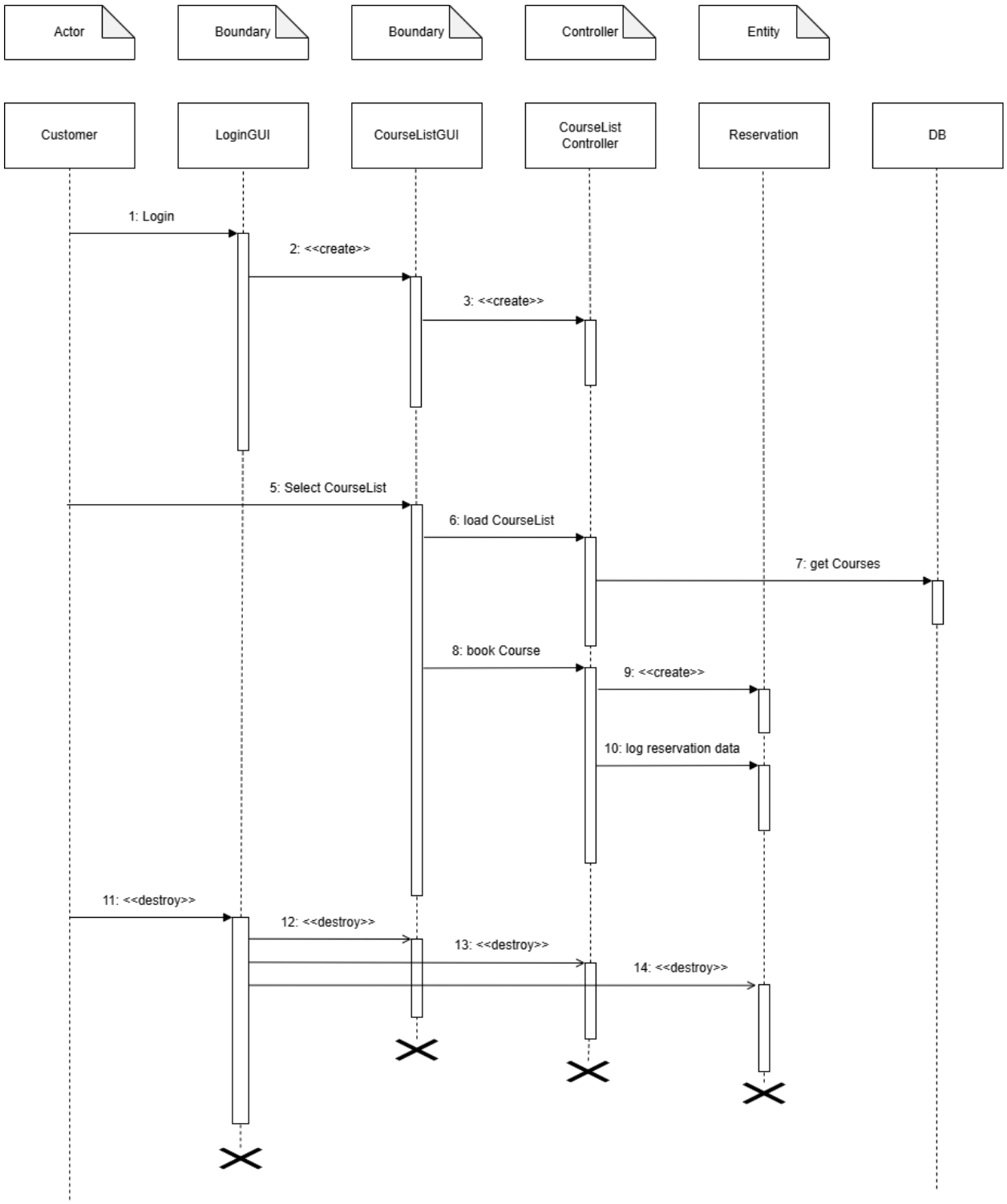

Login*
Enter Username and Password
Verify Username and Password
[Existing account]
[Account does not exist]
Login
Reject login
Show Homepage
Show error message

3.4 Sequence Diagram

UC: Renew Subscription

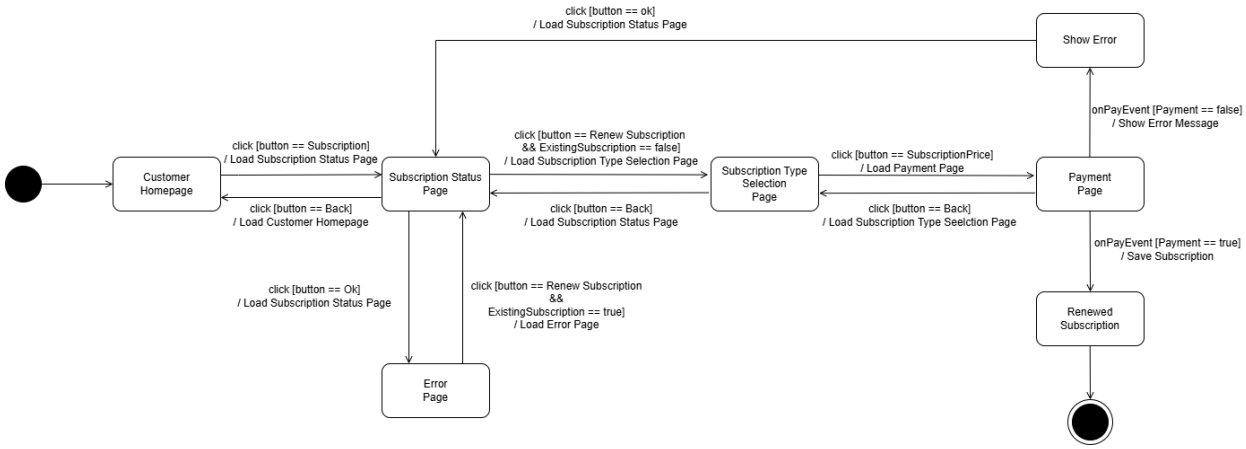


UC: Book an Activity

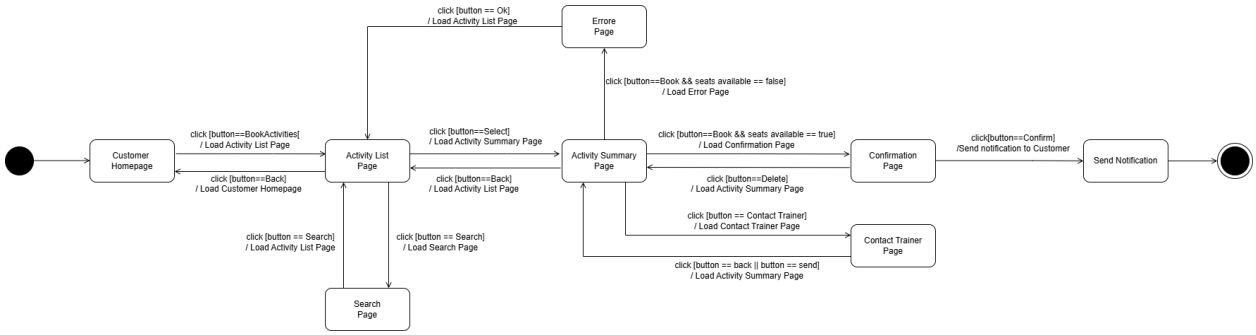


3.5 State Diagram

UC - Renew Subscription



UC - Book an Activity



4. Testing

Prevention of multiple active subscriptions - Chiara

We tested the subscription system, verifying that a user cannot activate a new one if they already have an active one. This test ensures the logical integrity of the system and prevents unauthorized subscription overlapping.

Uniqueness of the generated code - Chiara

We verified that each code generated (e.g., for QR codes or login) is unique, even in subsequent calls, ensuring the integrity and reliability of the identification system.

Restricting access to services for unauthenticated users - Chiara

We have tested whether an unauthenticated user can't perform confidential operations, such as creating a subscription, while respecting the system's security and authentication requirements.

Activity duplication control - Simone

The system was tested to prevent the creation of duplicate activities with the same name, date, and start time, ensuring the uniqueness of events in the calendar.

Management of maximum capacity activities - Simone

We verified that the system prevents bookings for an activity when the maximum number of participants has been reached.

Access to personalized cards - Simone

We verified that a customer cannot access or view training cards assigned to other customers, thus ensuring proper separation of user data.

5. Repository GitHub

https://github.com/SimoX26/GymWizard.git

6. SonarCloud

https://sonarcloud.io/project/overview?id=SimoX26_GymWizard

7. Presentation Video

Chiara:

PRESENTAZIONE SOFTWARE GYMWIZARD - CHIARA CARUMANI - YouTube